

Database Benchmark Summary

PostgreSQL vs ClickHouse - same data, identical benchmark runner, warmup and measured runs

What we did

Both databases, PostgreSQL and ClickHouse, were placed on the same DB machine: VM2.
The benchmark runner executed from VM1.

For each query and database, we ran 2 warmup runs first, then 10 measured runs.
The measured runs were used to calculate mean, median, min, max, standard deviation, and p99 runtime.

All benchmark executions succeeded, and row counts matched between PostgreSQL and ClickHouse for each query.

Bottom line

ClickHouse was faster on every tested query. The largest median gap was Q3, where ClickHouse was about 7.22x faster. Q2 was also about 6.06x faster. Q1 had a smaller but still clear advantage of about 1.35x.

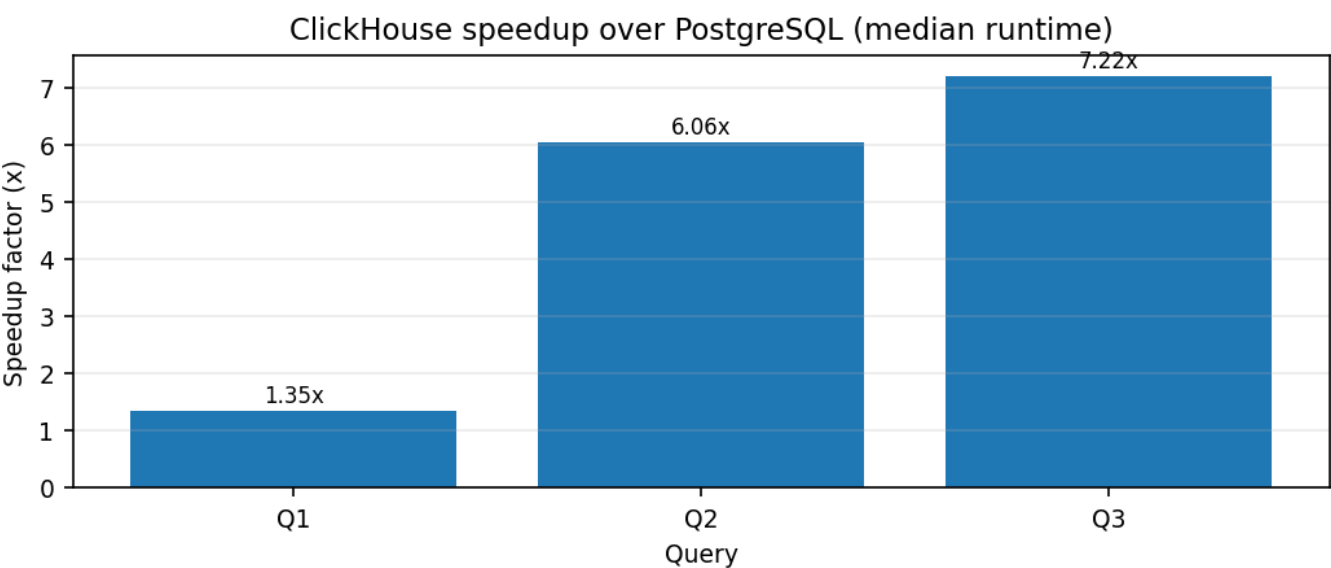
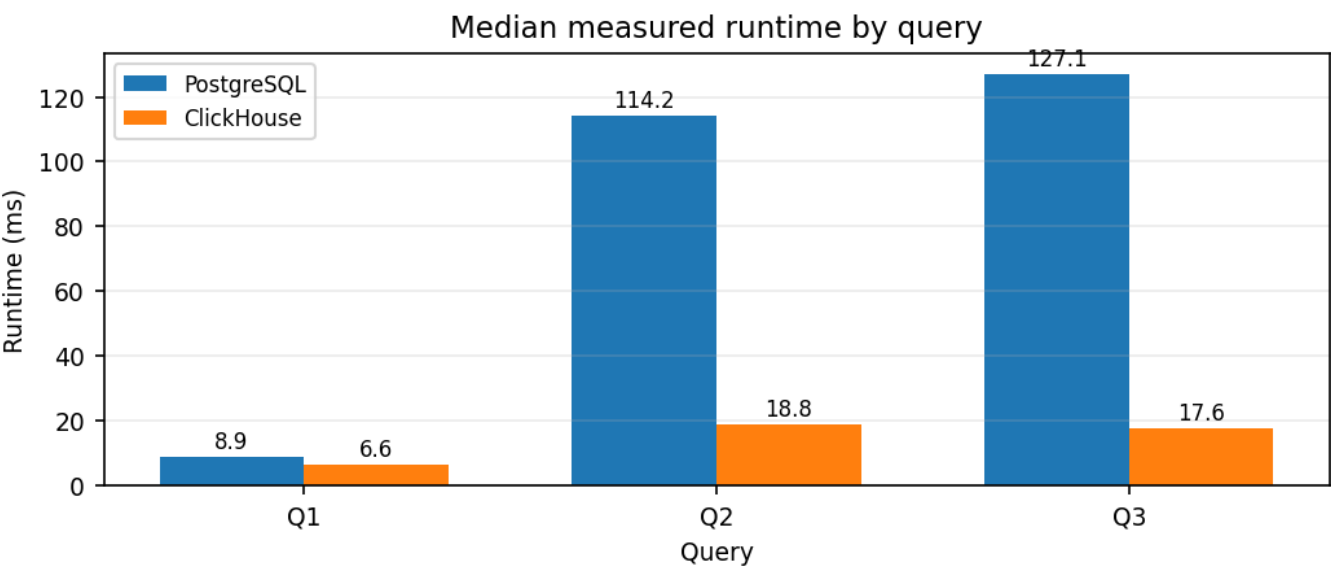
For Q3, ClickHouse had one runtime spike. Because of that, the median is more representative than the mean for that query.

Speedup summary

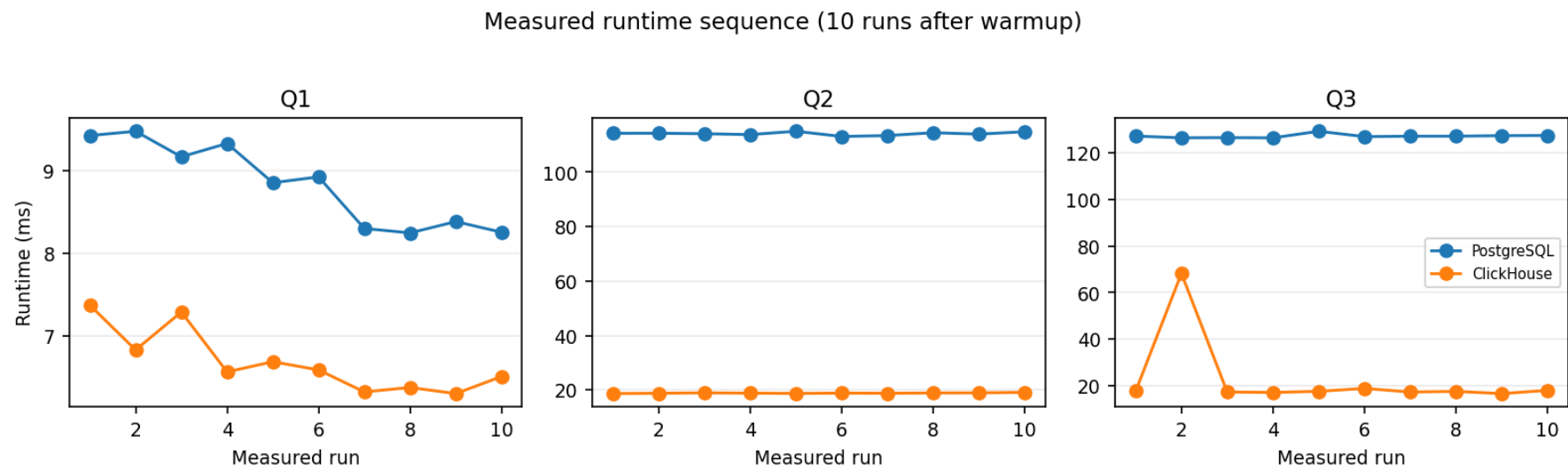
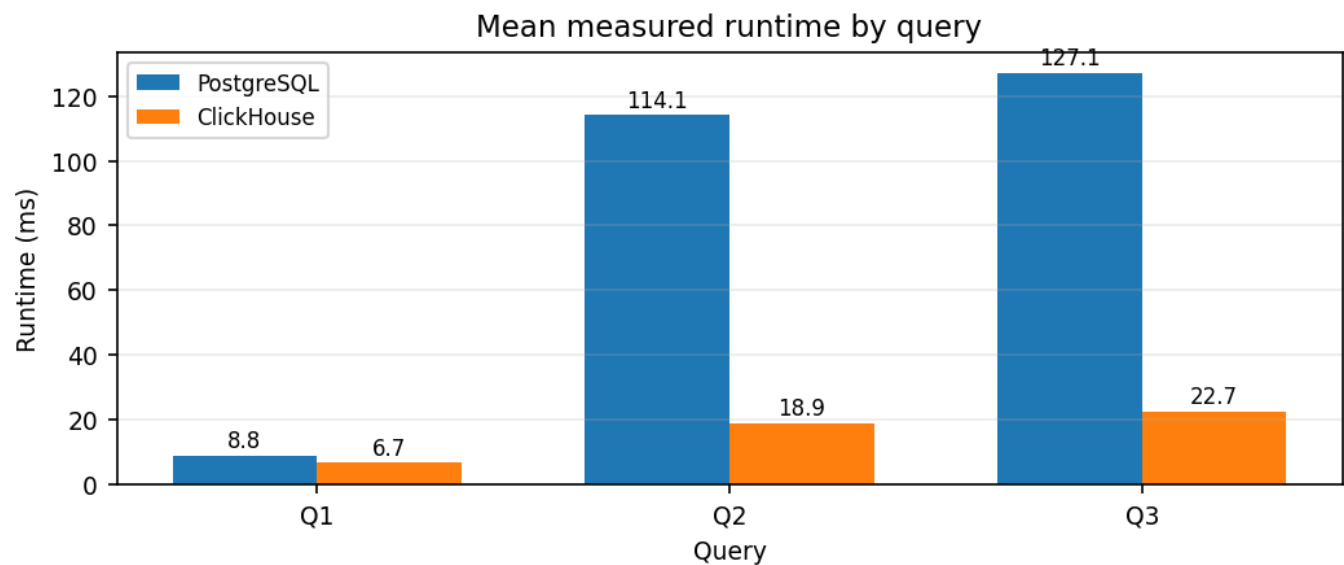
Query	Winner	Mean speedup	Median speedup	Comment
Q1	ClickHouse	1.32x	1.35x	Smallest gap, but ClickHouse is still faster.
Q2	ClickHouse	6.05x	6.06x	Largest mean speedup; very stable timings.
Q3	ClickHouse	5.61x	7.22x	Median is best because of one ClickHouse runtime spike.

Runtime graphs

All values in the charts are calculated from the 10 measured runs only, after the two warmup runs.



Runtime graphs - continued



Measured runtime statistics

Values are shown in milliseconds. Statistics are calculated from the 10 measured runs only, after the two warmup runs.

Query	Database	N	Mean	Median	Min	Max	Std	p99
Q1	PostgreSQL	10	8.836	8.889	8.245	9.478	0.506	9.474
Q1	ClickHouse	10	6.679	6.571	6.296	7.369	0.379	7.362
Q2	PostgreSQL	10	114.094	114.158	113.075	114.973	0.591	114.958
Q2	ClickHouse	10	18.857	18.845	18.665	19.126	0.130	19.109
Q3	PostgreSQL	10	127.124	127.082	126.360	129.183	0.815	129.020
Q3	ClickHouse	10	22.659	17.612	16.655	68.218	16.018	63.778

Query map and validation details

ID	Query file	Returned rows
Q1	q01_private_home_villa_weekend_long_calls.sql	4796
Q2	q02_top_2_longest_calls_by_zip_group_and_day_type.sql	8
Q3	q03_monthly_duration_metric_with_top_borough.sql	13

Execution details

Runner VM	DB VM	Warmup runs	Measured runs
VM1	VM2	2	10

Interpretation

The comparison is fair for a basic benchmark because both databases ran on the same DB VM and used the same warmup and measurement pattern. PostgreSQL was consistent, while ClickHouse was mostly consistent except for the Q3 spike. For that reason, median runtime is the best primary comparison metric.

This report is based on benchmark_summary.csv, benchmark_results.csv, and benchmark_notes.txt.